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(54) **MANUFACTURE OF STRAINED
FERMENTED DAIRY PRODUCTS**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2003/0196957 A1* 10/2003 Henningfield **C13K 5/00**
210/651
2013/0121976 A1* 5/2013 Montserrat Carreras
A61K 35/744
424/93.44
2014/0335226 A1* 11/2014 Bell **A23C 21/10**
426/576

FOREIGN PATENT DOCUMENTS

EP 0 402 450 A1 12/1990
WO 2013/153414 A1 10/2013
WO 2014/114970 A1 7/2014

OTHER PUBLICATIONS

NPL *Streptococcus thermophilus* (Retrieved on Jan. 4, 2023). [This
is also Google Scholar Search itself which is used to use as
evidentiary reference. (Year: 2022).*
International Search Report and Written Opinion from International
Application No. PCT/EP2016/059838 dated Jul. 20, 2016.

* cited by examiner

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(57) **ABSTRACT**

The invention relates to the manufacture of strained fer-
mented dairy products. The invention allows improvements
in the use of the materials and by-products as well as in the
properties of the product obtained. A lactic acid bacteria
having a low lactose metabolism capacity in acid whey is
used.

11 Claims, No Drawings

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**MANUFACTURE OF STRAINED
FERMENTED DAIRY PRODUCTS****RELATED APPLICATIONS**

This application is a national phase filing under 35 USC 371 of International Application No. PCT/EP2016/059838, filed on May 3, 2016, which claims priority of U.S. Provisional Application No. 62/158,390, filed on May 7, 2015, U.S. Provisional Application No. 62/158,386, filed May 7, 2015, U.S. Provisional Application No. 62/159,425, filed May 11, 2015, and U.S. Provisional Application No. 62/159,423, filed May 11, 2015, the entirety of which are incorporated herein by reference for all purposes.

The invention relates to the manufacture of strained fermented dairy products. The invention allows improvements in the use of the materials and by-products as well as in the properties of the product obtained.

Strained fermented dairy products, such as strained yogurts, are products obtained by a process involving a fermentation of a dairy material with lactic acid bacteria, and then a separation step, wherein on one hand a concentrated strained product is obtained, and on another hand an acid whey by-product is obtained. As production and consumption of strained products increases, the production of acid whey by-product also increases. The acid whey by-product however finds low usage, and large quantities are to be disposed of, preferably in a nature-friendly fashion, which can require costly treatments. Acid-whey comprises compounds that can be used, such as lactose. Lactose can be for example extracted and used in various applications. Such usage of lactose is however economically challenging: the less lactose the acid whey by-product comprises, the less economically viable the extraction and/or usage thereof is.

Indeed disposal of acid whey is not recommended and lactose valorization thereof is challenging if the lactose content is too low. The lactose content in acid whey has been found to decrease upon storage. Obviating this and maintaining a high level of lactose for further valorization can imply significant processing investments and/or operating costs. One solution can be to extract lactose directly after whey separation without transportation to another extraction site having the required equipment. This requires specific investments on the strained fermented product and acid whey by-product production site whereas capacity can be available on other sites. Such a solution lacks flexibility. Another solution can be to freeze the acid whey, to stabilize the lactose content between the recovery (by separation) and the extraction, for example during transportation. Such a solution requires much energy and/or specific transportation equipments. Here also the costs and/or the impacts on nature are not interesting.

There is a need in processes for manufacturing strained fermented dairy products, such as strained yogurts, that provides good products as well as improved possibilities of managing the acid whey by-product, for example allowing an improved valorization of the lactose contained therein.

The invention addresses at least one of the needs or problems above with a process for the manufacture of a strained fermented dairy product, comprising at least the following steps:

- a) heat treatment of a dairy material comprising lactose,
- b) fermentation with at least one lactic acid bacteria,
- c) separation to obtain a strained fermented dairy product and an acid whey by-product comprising lactose,
- d) optionally smoothing the fermented dairy product,
- e) optionally at least one cooling step,

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wherein the at least one lactic acid bacteria has a low lactose metabolization capacity in acid whey.

The invention also concerns the strained fermented dairy product that can be obtained by the process. The invention also concerns products comprising the strained fermented dairy product and the uses of the strained fermented dairy product.

Definitions

The term “acid whey” is used herein to describe a by-product of the separation step. The term “acid whey” also encompasses further processed compositions (e.g. filtered acid whey, neutralized acid whey and refined acid whey).

In the present application the lactose metabolization capacity in acid whey refers to the capacity of a lactic acid bacteria to consume lactose in acid whey. The metabolization capacity is typically measured on an acid whey composition having:

- from 0.0% to 0.4% by weight of protein,
- from 2.8% to 4.7% by weight of lactose,
- from 92.0% to 95% by weight of water,
- from 0.00% to 0.10% by weight of fat, and
- a pH of from 3.80 to 4.65.

The metabolization capacity is preferably determined on an acid whey composition having:

- 0.4% by weight of protein, preferably of whey protein,
- from 2.8% to 4.7% by weight of lactose,
- from 94.3% by weight of water,
- from 0.0% by weight of fat, and
- a pH of 4.5.

In the present application a low lactose metabolization capacity refers to a lactose loss of lower than 15%, preferably lower than 12%, preferably lower than 10%, preferably lower than 8%, preferably lower than 7%, after storage during 7 days at 32° C.

In the present application the lactose stability refers to the lactose conservation, as opposed to the lactose loss, after a storage, preferably of 7 days at 32° C.

Dairy Material

The invention involves processing a dairy material. The dairy material is typically comprised of milk and/or ingredients obtained from milk. It is also referred to as a “milk-based composition”. Herein milk encompasses animal milk, such as cow’s milk, and also substitutes to animal milk, such as vegetal milk, such as soy milk, rice milk, etc

Milk-based compositions useful in such products and/or processes are known by the one skilled in the art of dairy products, preferably of fermented dairy products. Herein a milk-based composition encompasses a composition with milk or milk fractions, and compositions obtained by mixing several previously separated milk fractions. Some water or some additives can be added to said milk, milk fractions and mixtures. Preferably the milk is an animal milk, for example cow’s milk. Some alternative animal milks can be used, such as sheep milk or goat milk.

The milk-based composition can typically comprise ingredients selected from the group consisting of milk, half skimmed milk, skimmed milk, milk powder, skimmed milk powder, milk concentrate, skim milk concentrate, milk proteins, cream, buttermilk and mixtures thereof. Some water or additives can be mixed therewith. Examples of additives that can be added include sugar, sweeteners different from sugar, fibers, and texture modifiers.

The milk-based composition can typically have a fat content of from 0.0% to 5.0% by weight, for example of from 0.0% to 1.0% or from 1.0% to 2.0% or from 2.0% to

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3.0% or from 3.0% to 4.0% or from 4.0% to 5.0%. The “fat content” of a composition corresponds to the weight of the fat components present in the composition relatively to the total weight of the composition. The fat content is expressed as a weight percentage. The fat content can be measured by the Weibull-Berntrop gravimetric method described in the standard NF ISO 8262-3. Usually the fat content is known for all the ingredients used to prepare the composition, and the fat content of the product can be calculated from these data.

The milk-based composition can typically have a protein content of from 2.0% to 6.0% by weight, for example of from 2.0% to 3.0% or from 3.0% to 4.0% or from 4.0% to 5.0% or from 5.0% to 6.0%. The “protein content” of a composition corresponds to the weight of the proteins present in the composition relatively to the total weight of the composition. The protein content is expressed as a weight percentage. The protein content can be measured by Kjeldahl analysis (NF EN ISO 8968-1) as the reference method for the determination of the protein content of dairy products based on measurement of total nitrogen. Nitrogen is multiplied by a factor, typically 6.38, to express the results as total protein. The method is described in both AOAC Method 991.20 (1) and international Dairy Federation Standard (IDF) 2013:1993. Usually the total protein content is known for all the ingredients used to prepare the product, and total protein content is calculated from these data.

The dairy material, also referred to as milk-based composition, comprises lactose. The amount of lactose can be typically of from 3.80% to 5.00% by weight.

In one embodiment the dairy material has the following contents (% by weight):

- from 3.0% to 3.5% of milk protein
- from 0.0% to 3.5% of fat
- from 3.80% to 5.00% of lactose.

The pH of the milk can for example be of from 6.60 to 7.00. The dry matter of the milk can be for example of from 6.8% to 13.0%. In one embodiment the milk is a low-fat milk comprising less than 2.0% fat, preferably less than 1.0% fat, preferably less than 0.5% fat. The milk can be for example a skimmed milk.

The ingredients of the milk-based composition and/or the amounts thereof can be selected to have the amounts of proteins and/or fat and/or lactose mentioned above.

Bacteria
The invention involves a fermentation with lactic acid bacteria. Such a step is known by the one skilled in the art. Appropriate lactic acid bacteria are known by the one skilled in the art. It is mentioned that lactic acid bacteria are often referred to as ferments or cultures or starters. Examples of lactic acid bacteria that can be used for the fermentation include:

- Lactobacilli*, for example *Lactobacillus acidophilus*, *Lactobacillus casei*, *Lactobacillus plantarum*, *Lactobacillus reuteri*, *Lactobacillus johnsonii*, *Lactobacillus helveticus*, *Lactobacillus brevis*, *Lactobacillus rhamnosus*, *Streptococci*, for example *Streptococcus thermophilus*, *Bifidobacteria*, for example *Bifidobacterium bifidum*, *Bifidobacterium longum*, *Bifidobacterium breve*, *Bifidobacterium animalis*,
- Lactococci*, for example *Lactococcus lactis*,
- Propionibacterium* such as *Propionibacterium freudenreichii*, *Propionibacterium freudenreichii ssp shermanii*, *Propionibacterium acidipropionici*, *Propionibacterium thoenii*,

mixtures or association thereof.

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The lactic acid bacteria preferably comprise, preferably essentially consist of, preferably consist of, *Lactobacillus delbrueckii ssp. bulgaricus* (i.e. *Lactobacillus bulgaricus*) and *Streptococcus salivarius ssp. thermophilus* (i.e. *Streptococcus thermophilus*) bacteria. The lactic acid bacteria used in the invention typically comprise an association of *Streptococcus thermophilus* and *Lactobacillus bulgaricus* bacteria. This association is known and often referred to as a yogurt symbiosis.

In some particular embodiments the lactic acid bacteria might comprise probiotic bacteria. Probiotic bacteria are known by the one skilled in the art. Examples of probiotic bacteria include some *Bifidobacteria* and *Lactobacilli*, such as *Bifidobacterium brevis*, *Bifidobacterium animalis animalis*, *Bifidobacterium animalis lactis*, *Bifidobacterium infantis*, *Bifidobacterium longum*, *Lactobacillus helveticus*, *Lactobacillus casei*, *Lactobacillus casei paracasei*, *Lactobacillus acidophilus*, *Lactobacillus rhamnosus*, *Lactobacillus plantarum*, *Lactobacillus reuteri*, *Lactobacillus delbrueckii subsp. bulgaricus*, *Lactobacillus delbrueckii subsp. lactis*, *Lactobacillus brevis* and *Lactobacillus fermentum*.

In one embodiment the lactic acid bacteria do not comprise *Bifidobacteria*. In one embodiment the lactic acid bacteria do not comprise *Lactobacillus acidophilus* bacteria. In one embodiment the lactic acid bacteria do not comprise *Bifidobacteria* and do not comprise *Lactobacillus acidophilus* bacteria.

The lactic acid bacteria can be introduced in any appropriate form, for example in a spray-dried form or in a frozen form. The introduction of the lactic acid bacteria in the dairy material is also referred to as an inoculation.

The invention involves using at least one lactic acid bacteria that has a low lactose metabolism capacity in acid whey, as defined and/or described above. Thus within the lactic acid bacteria mentioned above, at least one bacteria strain is to exhibit a low lactose metabolism in acid whey.

In one embodiment the at least one lactic acid bacteria having a low lactose metabolism capacity in acid comprise a *Lactobacillus bulgaricus* strain. Examples of such *Lactobacillus bulgaricus* strains include *Lactobacillus bulgaricus* strain CNCM I-2787 (deposited according to the Budapest treaty with the Collection Nationale de Cultures de Microorganismes as the international depositary authority, on Jan. 24, 2002 under number I-2787).

In one embodiment the fermentation step b) is carried out with a culture comprising, preferably essentially consisting of, preferably consisting of, at least one *Streptococcus thermophilus* strain, and at least one *Lactobacillus bulgaricus* strain.

The *Streptococcus thermophilus* bacteria preferably comprise:

- Streptococcus thermophilus* strain CNCM I-2784 (deposited according to the Budapest treaty with the Collection Nationale de Cultures de Microorganismes as the international depositary authority, on Jan. 24, 2002 under number I-2784),
- Streptococcus thermophilus* strain CNCM I-2835 (deposited according to the Budapest treaty with the Collection Nationale de Cultures de Microorganismes as the international depositary authority, on Apr. 4, 2002 under number I-2835), and/or
- Streptococcus thermophilus* strain CNCM I-2773 (deposited according to the Budapest treaty with the Collection Nationale de Cultures de Microorganismes as the international depositary authority, on Jan. 24, 2002 under number I-2773),

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The *Lactobacillus thermophilus* bacteria preferably comprise:

Lactobacillus bulgaricus strain CNCM I-2787 (deposited according to the Budapest treaty with the Collection Nationale de Cultures de Microorganismes as the international depositary authority, on Jan. 24, 2002 under number I-2787).

Herein, the following references are also used:

DN-001640 to designate *Streptococcus thermophilus* strain CNCM I-2784,

DN-001336 to designate *Streptococcus thermophilus* strain CNCM I-2835,

DN-001236 to designate *Streptococcus thermophilus* strain CNCM I-2773, and

DN-100290 to designate *Lactobacillus bulgaricus* strain CNCM I-2787.

Step a)—Heat Treatment

The process of the invention involves heat treating the dairy material in a step a). Such heat treatments are known by the one skilled in the art, for example as pasteurization or sterilization. They allow eliminating parasite micro-organisms. They can be performed in conventional heat exchangers, such as tubes or plates heat exchangers. The heat treatment can be for example performed at a temperature of from 80° C. to 99° C., preferably 85° C. to 95° C., for example during from 1 minute to 15 minutes.

It is mentioned that the process can comprise a homogenization step before or after the heat treatment step, preferably at a pressure of from 20 bars to 300 bars, in particular from 50 bars to 250 bars.

It is mentioned after the heat treatment the dairy material is typically cooled down to a fermentation temperature.

Step b)—Fermentation

The process of the invention involves a fermentation step with at least one lactic acid bacteria. In this step the dairy material is inoculated with the lactic acid bacteria, and the mixture is then allowed to ferment at a fermentation temperature. Such inoculation and fermentation operations are known by the one skilled in the art.

During fermentation, the lactic acid bacteria produce lactic acid and thus cause a pH decrease. With the pH decreasing proteins coagulate to form a curd, typically at a breaking pH.

The fermentation temperature can be of from 30° C. to 45° C., preferably from 35° C. to 40° C., with a pH decrease to a breaking pH at which proteins coagulate to form a curd.

The breaking pH is preferably of from 3.50 to 5.50, preferably of from 4.0 to 5.0, preferably from higher than 4.5 to 5.0.

Step c)—Separation

The process of the invention involves a separation step. In this step an acid whey composition is separated from the curd resulting from the proteins coagulation. Thus one obtains:

- a fermented dairy product, typically comprising the proteins coagulum, referred to as a strained fermented dairy product, and
- an acid whey by-product.

Such separation steps are known by the one skilled in art, for example in processes of making "greek yogurts". The separation can for example be carried out by reverse osmosis, ultrafiltration, or centrifugal separation. The separation step can be performed for example at a temperature of from 30° C. to 45° C.

The acid whey by-product comprises lactose, for example as further described below. In one embodiment an amount of

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from 65% to 90% by weight, preferably from 70% to 85%, with reference to the amount of dairy material, of acid-whey by-product is recovered.

The strained fermented dairy product comprises a high amount of proteins and is suitable and valuable of consumption. It is also referred to herein as "White Mass".

Step d)—Smoothing

The process of the invention can comprise a step wherein the strained fermented dairy product undergoes a smoothing step. Such steps typically, involving some agitation and/or shear, and are known by the one skilled in the art. The smoothing step can be performed for example by agitation, or by static or dynamic smoothing. In one embodiment the smoothing is a dynamic smoothing, performed with a rotor stator mixer. An example of such an equipment is given in the patent application WO2007/095969. In the context of the invention, "rotor stator mixer" means an equipment in which the product goes through clogged rings, a part of the rings being static, the remaining part being in rotation at a set speed. This system of clogged rings partly static or in rotation applies a defined shearing to the product. Preferably, the rotor stator mixer comprises a ring shaped rotor and a ring shaped stator, each ring of the rotor and the stator being provided with radial slots having a given width, comprising adjusting the rotational speed of the rotor to adjust the peripheral velocity. The rotor may be operated so that the peripheral velocity is between 2 m/s and 13 m/s, in particular between 3 m/s and 5 m/s and more particularly between 3.6 m/s and 4 m/s. For example the process can comprise a dynamic smoothing step, preferably performed with a rotor stator mixer, preferably at a temperature of from 30° C. to 45° C.

Temperatures

In a preferred embodiment:

the heat treatment step a) is performed at a temperature of from 80° C. to 99° C., preferably 85° C. to 95° C.,

the fermentation step b) is performed at a temperature of from 30° C. to 45° C., and or

the separation step is performed at a temperature of from 30° C. to 45° C.

It is mentioned that the process of the invention can comprise at least one cooling step. For example the process can involve a cooling between the heat treatment step and the fermentation step. The process can involve a cooling step performed on the strained fermented dairy product, to reach a storage temperature, for example a chilled temperature of from 1° C. to 10° C., for example 4° C. The process can involve a cooling step performed on acid whey by-product, to reach a storage temperature, for example a room temperature. In one embodiment the process comprises a cooling step e1) of the fermented dairy product, to a temperature of from 4° C. to 10° C. In one embodiment the process comprises a cooling step e2) of the acid whey by-product to a room temperature, preferably to from 15° C. to 25° C.

In one embodiment the process of the invention comprises a heat treatment step, typically a temperature increase step, at the end of the fermentation and before the separation, referred to as thermoshocking step. This step is typically performed by raising the temperature to a temperature from 50° C. to 75° C., preferably from 50° C. to 60° C. Such a thermoshocking step can contribute to stabilizing the organoleptic properties of the strained dairy fermented product. Alternatively, a heat treatment can be performed after the separation step on the acid whey by product with similar increase in temperature. It has been surprisingly found that such a thermoshocking step can also contribute to stabilizing

the amount of lactose in the acid whey by-product. It is believed that at least a part of the lactic acid bacteria remains alive after such a treatment.

In one embodiment the process involves the following phases:

Fermentation→Temperature increase (Thermoshocking)
→Separation→cooling of strained fermented dairy product and of acid whey by-product.

In one embodiment the process involves the following phases:

Fermentation→Separation→Cooling of strained fermented dairy product and temperature increase (Thermoshocking) of acid whey by-product→cooling of acid whey by-product.

These embodiments are found to be efficient from an energy management point of view as allowing an increase of temperature (Thermoshocking) from a fermentation or separation temperature typically of from 30° C. to 45° C. to a temperature of from 50° C. to 75° C. Such embodiments consume less heating and/or cooling energy than embodiment wherein the acid whey by-product would be cooled and then significantly heat-treated for example at a pasteurization or sterilization temperature.

Acid Whey By-Product

The acid whey by-product is recovered at the separation step. Preferably the acid whey by-product:

has a lactose content of at least 2.80% by weight, preferably at least 3.00%, preferably at least 3.2%, preferably at least 3.50%, preferably at least 4.00%, and comprises the at least one lactic acid bacteria having a low lactose metabolism capacity in acid whey.

The acid whey by-product typically comprises water, for example in an amount of higher than 90% by weight. The acid whey by-product typically comprises the at least one lactic acid bacteria having the low lactose metabolism capacity, preferably in an alive state. The lactose stability is preferably of higher than 85%, preferably higher than 88%, preferably higher than 92%, preferably higher than 93%, preferably higher than 94%, preferably higher than 95%, preferably higher than 96%, preferably higher than 97%, preferably higher than 98%, preferably higher than 99%, during a storage of 7 days at 32° C., preferably of storage of 7 days from a production at day 0.

In one embodiment the lactose content in the acid whey by-product is up to 6.00% by weight, preferably up to 5.00%.

The acid whey by-product preferably has the following contents (% by weight), preferably just after collection from separation:

from 0.0% to 0.4% of protein,
from 2.80% to 4.70% of lactose, and
from 92.0% to 95% of water.

The pH of the acid whey by-product can for example be of from 3.50 to 4.70, preferably from 3.80 to 4.65. The acid whey by-product is typically substantially free of fat.

It is mentioned that the amount of lactose in the acid whey by-product is typically lower than the initial amount of lactose in the in dairy material. Typically the amount of lactose in the acid whey by-product is of at least 5% less than in the dairy material, preferably at least 10% less.

The acid whey typically comprises the at least one lactic acid bacteria having a low lactose metabolism capacity in the acid whey. The acid whey by-product can comprise other lactic acid bacteria used in the fermentation step. It is mentioned that the lactic acid bacteria comprised in the acid

whey by-product are typically alive, particularly the at least one lactic acid bacteria having a low lactose metabolism capacity in the acid whey.

In some embodiments, the acid whey by-product is cooled after the separation step. In some embodiments, the acid whey is cooled to a room temperature or below a room temperature. In some embodiment a thermoshocking temperature increase or heat treatment is performed between the separation and the cooling.

The acid whey by-product is then typically used for lactose recovery (for example extraction by isolation and/or purification) or other applications wherein presence of lactose is valuable.

Advantageously the acid whey by-product does not undergo a heat treatment step after separation at a temperature that might kill the bacteria comprised therein, for example at a temperature of above 75° C. The process according to the invention allows avoiding such a heat treatment step and thus allows energy savings and/or simplification.

In some embodiments, the process extends the lactose shelf-life in the acid whey by-product by 3 days or more. In some embodiments, the process extends the lactose shelf-life in the acid whey by-product by 7 days or more. In some embodiments, the method extends the lactose shelf-life in the acid whey by-product by 15 days or more. In some embodiments, the process extends the lactose shelf-life in the acid whey by-product by 3 days to 15 days. In some embodiments, the process extends the lactose shelf-life in the acid whey by-product by 3 days to 7 days. In some embodiments, the process extends the lactose shelf-life in the acid whey by-product by 7 days to 15 days. The extension of shelf-life is typically considered with reference to acid whey by-products that do not comprises the at least one lactic acid bacteria that has a low lactose metabolism capacity, preferably in an alive state.

The stabilization of the lactose content of the acid whey by-product (or any carbohydrate derived from it such as glucose or galactose) allows maximizing its value for various applications. Examples valorizations include:

Isolation and purification of lactose to produce crystalline lactose. Crystalline lactose has value for food applications (such as infant milk formula) and pharmaceutical applications as a filler in various tablet formulations,

Transformation of lactose into other carbohydrates through enzymatic treatments (lactases, invertases) to produce glucose, galactose or other sugar of interest,

Transformation of lactose into polysaccharides such as Galacto-Oligo-Saccharides (GOS) through enzymatic treatment (reverse-lactase), than can be used as a fiber or a functional prebiotic in food applications,

Utilization of the lactose-rich acidic (or neutralized) whey as a medium to grow biomass with micro-organisms of interest, such as yeast, for human or animal nutrition,

Utilization of the lactose-rich whey to grow biomass such as methane-producing micro-organisms for energy production (biodigestion)

Fermentation with yeasts, for example with yeasts belonging to the genus *Kluyveromyces* that have a unique industrial application as they are capable of fermenting lactose for ethanol production. Surplus lactose from the whey by-product is a potential source of alternative energy.

Strained Fermented Dairy Product

The strained fermented dairy product is recovered at the separation step. As much water has been removed as part of the acid whey by-product, the strained fermented dairy

product comprises high amounts of proteins, especially of casein. Thus the product is also referred to as "White Mass".

The strained fermented dairy product comprises lactic acid bacteria, wherein the lactic acid bacteria comprise at least one lactic acid bacteria having a low lactose metabolism capacity. All the features mentioned above about lactic acid bacteria used in the fermentation step apply to the lactic acid bacteria comprised in the strained dairy fermented product.

Thus in the strained fermented dairy product the at least one lactic acid bacteria preferably comprises a *Lactobacillus bulgaricus* strain.

In one embodiment the lactic acid bacteria comprise, preferably essentially consist of, preferably consist of, at least one *Streptococcus thermophilus* strain, and at least one *Lactobacillus bulgaricus* strain.

In one embodiment the *Lactobacillus bulgaricus* strain is strain CNCM I-2787.

In one embodiment:

the *Streptococcus thermophilus* strain comprise at least one *Streptococcus thermophilus* strain selected from the group consisting of strain CNCM I-2784, strain CNCM I-2835, strain CNCM I-2773 and mixtures or associations thereof, and

the *Lactobacillus bulgaricus* strain is strain CNCM I-2787.

The strained fermented dairy product preferably has the following contents (% by weight):

from 8.5% to 11.0% of milk protein

from 0.0% to 8.0% of fat, for example from 0.0% to 3.5% or from 3.5% to 8.0%

from 0.00% to 4.20% of lactose, for example from 2.80% to 4.20%

The pH of the strained fermented dairy product can for example be of from 3.80 to 4.65.

It is mentioned that the amount of lactose in the strained fermented product is typically lower than the initial amount of lactose in the dairy material. Typically the amount of lactose in strained fermented dairy product is of at least 5% less than in the dairy material, preferably at least 10% less.

Final Product or Composition

The strained fermented dairy product is typically a final composition ready for consumption, or a part thereof. Thus the strained fermented dairy product can be used directly or associated or mixed with intermediate preparations such as fruit preparations or syrups, sauces such as chocolate or caramel sauces, or addition of organoleptic modifiers such as sweeteners, sugar or flavors. Such associations or mixtures and such preparations are known by the one skilled in the art. The amount by weight of intermediate preparations can be for example of from 1% to 90%, with reference to the total weight of final composition, for example from 5% to 25% by weight for solid compositions products or from 50% to 90% by weight, preferably of fruit juice, for smoothies.

Examples of final products include:

Fruit On the Bottom (FOB) products, having a fruit preparation layer on the bottom of a container comprising the composition, and an upper layer of the white mass,

Plain products, typically the white mass,

Sauce on top, having the white mass layer on the bottom of a container comprising the composition and an upper sauce layer, for example a chocolate or caramel sauce,

Stirred products or drinks, for example smoothies, being a mixture of the white mass and of an intermediate

preparation such as a fruit juice or a fruit preparation or syrup. For such products, the mixture can be performed before a smoothing step.

The final composition is typically contained in a sealed container such as a packaging. The process can typically involve a step of conditioning the final composition in a container. The container is then typically sealed, for example with a cap or a lid. The container can be for example a container of 50 ml (or 50 g), to 1 L (or 1 kg), for example a container of 50 ml (or 50 g) to 80 ml (or 80 g), or 80 ml (or 80 g) to 100 ml (or 100 g), or 100 ml (or 100 g) to 125 ml (or 125 g), or 125 ml (or 125 g) to 150 ml (or 150 g), or 150 ml (or 150 g) to 200 ml (or 200 g), or 200 ml (or 200 g) to 250 ml (or 250 g), or 250 ml (or 250 g) to 300 ml (or 300 g), or 300 ml (or 300 g) to 500 ml (or 500 g), or 500 ml (or 500 g) to 750 ml (or 750 g), or 750 ml (or 750 g) to 1 L (or 1 kg).

The final composition can be stored, transported and/or distributed at a chilled temperature of 0° C. to 10° C., preferably of 4° C. to 10° C.

Use of the Final Product or Composition

The final composition or product is typically to be used as a food product. It is typically used by oral administration. One can typically eat or drink the composition by processing it from a container to the mouth, optionally with using a spoon or a straw.

Further details or advantages of the invention might appear in the following non limitative examples.

EXAMPLES

Example 1—Manufacture of Strained Fermented Dairy Products and Acid Whey By-Products

Strained fermented dairy products are manufactured at pilot scale with using the following ingredients:

Milk: Skimmed milk having 3.17% protein, 0% fat and 8.8% dry matter

Cultures:

Culture 1: Yo-Mix® 495, marketed by Dupont

Culture 2: Mixture of the following bacterial strains:

Streptococcus thermophilus

DN-001640, *Streptococcus thermophilus* DN-001336,

Streptococcus thermophilus

DN-001236, and *Lactobacillus bulgaricus*

DN-100290.

The procedure involves the following steps:

heat treatment of milk at a temperature of 95° C. during 6.5 minutes,

homogenization at a temperature of 60° C., at a pressure of 69 bars,

inoculation of milk at 40° C. with 0.02% by weight of culture,

fermentation at a temperature of 40° C. to reach a breaking pH of 4.65,

optionally: temperature increase ("fermented mix thermoshock") to a temperature of 59.5° C. during 2.5 minutes,

separation, at a temperature of 41.5° C., of 72% of whey, with a Westphalia KNA3 pilot scale centrifuge separator, to obtain:

A) a strained fermented dairy product, and

B) an acid whey by-product, and

dynamic smoothing, performed on the strained fermented dairy product.

Example 2—Acid-Whey

The acid whey is collected as aliquot to sterile specimen cups. Separate samples are collected:

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A “reference” sample which is aliquot and immediately frozen by placing it in a chamber to stop any lactose metabolism,

A “32° C. storage” sample which is aliquot and placed in a 32° C. chamber,

A “4° C. storage” sample which is aliquot and placed in a 4° C. chamber,

All acid whey samples are kept in their respective chambers for 7 days before they are frozen in the −4° C. chamber and then analyzed within 24 hours of chamber transfer.

Acid Whey Analysis

The lactose content and *Streptococcus* bacteria populations are analyzed (National Food Lab, Livermore, California).

Lactose analysis results are reported on tables 1 and 2 below:

Remaining lactose in the acid whey (g of lactose per 100 g of acid whey)

Streptococcus thermophilus count (CFU per g)

Lactose loss, compared to the “reference” sample:

Lactose loss=(sample value−reference reference)/reference value Here a negative value indicates a loss.

TABLE 1

	Culture 1	Culture 2
“32° C. Storage”	3.16	3.87
Remaining lactose (%)		
“32° C. Storage”	−22%	−6.5%
Lactose loss		

This shows that Culture 2 allows higher conservation of lactose in acid whey.

TABLE 2

		Remaining lactose (%)	<i>S. thermophilus</i> (CFU/g)	Lactose Loss
Culture 1	4° C. Storage	3.81	6.0×10^7	−6%
	32° C. Storage	3.16	9.6×10^6	−22%
Culture 2*	4° C. Storage	3.87	1.4×10^8	−3.5%
	32° C. Storage	3.76	8.0×10^3	−6.5%

*average on 2 productions

Under 32° C. chamber conditions, the highest level of biomass available is found with Culture 2. Interestingly and surprisingly, Culture 2 also has the lowest level of lactose at those conditions which means a lesser amount of biomass is needed to consume lactose. This shows that a culture selection can play a role in the stabilization of lactose in the acid whey.

Example 3—Strained Fermented Dairy Product

The strained fermented dairy products, also referred to as “White Mass” (WM), are processed as finished products. For plain products 6 oz of White Mass are conditioned in cups.

For Strawberry Fruit On the Bottom (FOB) products, 2 oz (25%) of a strawberry preparation and then 4 oz (75%) of White Mass are dosed in a cup.

The products obtained with Culture 1 and with Culture 2 are compared for overall balance by a trained panel, at D28 (28 days of storage at 4° C. after preparation) and D55 (55 days of storage at 4° C. after preparation). Most significant and important differences in attributes are reported on table 3 below.

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Overall balance attributes: Evaluation of the roundness of flavor, of the lack of spike from any tastes or flavor, and of the lack of overpowering notes.

TABLE 3

	Culture 2, with reference to Culture 1
Plain Product D28	More overall balance (score difference 0.6)
Plain Product D55	More overall balance (score difference 0.8)
Strawberry FOB Product D28 (stirred before tasting)	More overall balance (score difference 0.6)
Strawberry FOB Product D55 (stirred before tasting)	More overall balance (score difference 0.6)

These results show that the products obtained with Culture 2 have an improved overall balance, and that the difference is even higher for the White Mass part after an extended shelf-life of 55 days.

Example 4—Post-Acidification

The post-acidification of the white mass is evaluated by pH measurements at D0 (after preparation), and D7 (7 days of storage at 4° C. after preparation). The results are reported on table 4 below.

TABLE 4

	Plain Product with Culture 1	Plain Product with Culture 2
pH at D0	4.50	4.45
pH at D7	4.45	4.40
Loss of pH from D0 to D7	−1.1%	−1.1%

This shows that the post-acidification of the strained fermented dairy product is similar with Culture 1 (pH drop of 1.1%) and Culture 2 (pH drop of 1.1%). However, surprisingly, the lactose stability in the corresponding acid whey by-products is very different with Culture 1 (lactose loss of 22%) and Culture 2 (lactose loss of 6.5%), as shown on table 1 and table 2. This shows that the lactose metabolism capacity of the cultures in acid whey is not directly correlated to post-acidification capacity in the strained fermented dairy product.

The invention claimed is:

1. A process for manufacturing a strained fermented dairy product, comprising:

- heating a dairy material comprising lactose,
- fermenting with lactic acid bacteria comprises at least three of strain CNCM I-2784, strain CNCM I-2835, strain CNCM I-2773, and strain CNCM I-2787,
- separating to obtain a strained fermented dairy product and an acid whey by-product comprising lactose, wherein the acid whey by-product comprises:
 - 92.0% to 95% water by weight,
 - 2.80% to 4.7% lactose by weight,
 - 0.1% to 0.4% protein by weight,
 - at least one lactic acid having a low lactose metabolism capacity,
 - wherein the lactose has a stability that is higher than 85% during 7 days of storage at 32° C., and
- heating at a temperature of from 50° C. to 75° C. to thermoshock the acid whey by-product, wherein the heating to a temperature of from 50° C. to 75° C. stabilizes a plurality of organoleptic proper-

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ties associated with the strained fermented dairy product to form at least one valorization, wherein the heating to a temperature of from 50° C. to 75° C. preserves the lactic acid bacteria comprising the at least three of strain CNCM I-2784, strain CNCM I-2835, strain CNCM I-2773, and strain CNCM I-2787, wherein the heating to a temperature of from 50° C. to 75° C. extends a shelf-life of the acid whey by-product for 15 days and capable of oral administration.

2. The process according to claim 1, wherein the at least three of strains include the lactic acid bacteria is strain CNCM I-2787.

3. The process according to claim 1, wherein: the lactic acid bacteria comprises at least one *Streptococcus thermophilus* strain selected from the group consisting of strain CNCM I-2784, strain CNCM I-2835, strain CNCM I-2773 and mixtures or associations thereof, and strain CNCM I-2787.

4. The process according to claim 1, wherein: heating step a) is performed at a temperature of from 80° C. to 99° C., fermenting step b) is performed at a temperature of from 30° C. to 45° C., and/or

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separation step c) is performed at a separation temperature of from 30° C. to 45° C.

5. The process according to claim 1, comprising a homogenization step before or after the heating step a).

6. The process according to claim 1, wherein at step c) an amount of from 10% to 30% by weight, with reference to the amount of dairy material, of acid-whey by-product is recovered.

7. The process according to claim 1, further comprising a smoothing step e), wherein the smoothing step e) is a dynamic smoothing step including agitation and/or shear.

8. The process according to claim 7, comprising at least one cooling step f), wherein the at least one cooling step f) comprises:

15 f1) cooling the fermented dairy product, and/or f2) cooling the acid whey by-product.

9. The process according to claim 8, comprising the at least one cooling the step f1) wherein cooling is performed to a temperature of from 4° C. to 10° C.

20 10. The process according to claim 8, comprising the at least one cooling step f2) wherein cooling is performed to a temperature of from 15° C. to 25° C.

11. The process according to claim 1, wherein the heating step d) is performed at a temperature of from 50° C. to 60° C.

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